Supplementary material

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Table S1: Scene-level split of the training and evaluation data. Roles are pre-training, fine-tuning, validation, pixel-wise test and point-based test. No-data is the fraction of pixels outside the valid area or marked no-data; it is not the clear class. The 121 Proba-V scenes are a subset of the 331 Level-2A products of the ESA Proba-V Cloud Detection Round Robin, acquired on 21 March, 21 June, 21 September and 21 December 2014. Scene IDs and the overlap check are provided at `https://github.com/IPL-UV/fdr4vgt-cloud`.

Role	Sensor	Year	Scenes	Pixels	Points	No-data / outside
Pre-train	Proba-V	2014	1 387	19 620 892 143	–	–
Pre-train	Proba-V	2019	1 408	20 257 544 156	–	–
Pre-train	SPOT-VGT	1999	414	6 694 805 490	–	–
Pre-train	SPOT-VGT	2012	438	6 708 212 403	–	–
Fine-tune	Proba-V	2014	41	76 261 744	–	45.8%
Fine-tune	SPOT-VGT	1999	25	515 770 988	–	50.5%
Validation	Proba-V	2014	7	12 255 634	–	48.0%
Validation	SPOT-VGT	1999	5	55 265 191	–	47.0%
Test (pixel)	Proba-V	2014	24	52 056 426	–	46.1%
Test (pixel)	SPOT-VGT	1999	2	22 716 480	–	32.4%
Test (pixel)	SPOT-VGT	2012	6	104 009 641	–	39.7%
Test (points)	Proba-V	2014	49	–	7 654	–
Annotated, not used	SPOT-VGT	1999	7	73 816 702	–	43.0%

Fine-tuning, validation and test share no scene ID, and no manual test annotation was used at any training stage. Pre-training includes 45 of the 166 annotated scenes, all SPOT-VGT: 25 fine-tuning, 5 validation, 8 test and 7 unused. There they carry only legacy pseudo-labels, never the manual reference masks. These 45 scenes are 1.2% of the 3,647 pre-training scenes and 1.4% of the pre-training pixels. The Proba-V scenes, including the 49 point-based test scenes, are disjoint from pre-training, fine-tuning and validation.

Table S2: Paired scene-level comparison of Multi-FT PW-DL3 against the other strategies and the legacy baselines. Each row compares the same scenes: two-sided Wilcoxon signed-rank test, Holm correction within sensor and metric, bootstrap 95% confidence interval of the median difference, and number of scenes where Multi-FT is better.

(a) Proba-V, 49 point-based scenes, F_2.

Comparison	Median ΔF_2	95% CI	Scenes better	p (Holm)
Multi-FT vs. pre-training	+0.027	0.018 to 0.045	37/49	1×10^{-4}
Multi-FT vs. Proba-FT	0.000	−0.006 to 0.005	22/49	0.74
Multi-FT vs. SPOT-FT	+0.023	0.015 to 0.036	37/49	1×10^{-4}
Multi-FT vs. BS2	+0.028	0.007 to 0.041	34/49	9×10^{-4}
Multi-FT vs. BS1	+0.032	0.008 to 0.058	35/49	8×10^{-3}

(b) Proba-V, 49 point-based scenes, comparison against Proba-FT in the three metrics.

Metric	Median difference	95% CI	Scenes better	p (Holm)
F_2	0.000	−0.006 to 0.005	22/49	0.74
IoU	+0.028	0.000 to 0.043	30/49	2×10^{-3}
Kappa	+0.040	0.006 to 0.072	35/49	4×10^{-4}

(c) SPOT-VGT, 8 pixel-wise scenes, F_2.

Comparison	Median ΔF_2	95% CI	Scenes better	p (Holm)
Multi-FT vs. pre-training	+0.032	0.003 to 0.052	7/8	0.31
Multi-FT vs. SPOT-FT	+0.004	0.001 to 0.017	7/8	0.31
Multi-FT vs. Proba-FT	+0.006	−0.019 to 0.033	4/8	0.64
Multi-FT vs. BS2	+0.019	−0.001 to 0.058	6/8	0.31
Multi-FT vs. BS1	+0.088	0.040 to 0.148	7/8	0.078

With eight scenes the Wilcoxon test has little power: the smallest attainable two-sided p-value is 0.0078, and after Holm correction none of the SPOT-VGT comparisons reaches 0.05. The intervals and the direction of the differences are reported instead of a significance claim.

Table S3: Magnitude of the change in cloud fraction across the VGT2 to Proba-V transition (cut at 2014-01-01) at the two sites of Fig. 3. Values are means of the daily cloud-fraction series, and the change is after minus before. Intervals are block-bootstrap 95% intervals of the difference in means, with 30-day blocks and 5,000 replicates.

Site	Series	Days before	Days after	Mean before	Mean after	Change (95% CI)
SW Australia coast	Legacy (BS1/BS2)	5 204	2 009	0.443	0.483	−0.040 (−0.068 to −0.016)
SW Australia coast	Multi-FT PW-DL3	5 204	2 009	0.512	0.519	−0.007 (−0.036 to +0.018)
BC glacier/snow	Legacy (BS1/BS2)	5 421	2 052	0.804	0.723	+0.080 (+0.051 to +0.112)
BC glacier/snow	Multi-FT PW-DL3	5 414	2 069	0.776	0.717	+0.060 (+0.029 to +0.094)

The legacy record switches from BS1 to BS2 at the transition, so its change combines the sensor change with the change of algorithm. At the coastal site the change is about five times larger for the legacy record than for the proposed model. At the glacier site both records change, and the difference between them is smaller than at the coast.